Roi Bein

rb505@duke.edu | (864) 650-7422 | Website

EDUCATION

Duke University, Durham, NC — Expected May 2029
BSE - Mechanical Engineering
Certificate: Aerospace Engineering
GPA: 3.96

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Onshape, CFD, FEA
Manufacturing: CNC Mill, Lathe, Waterjet, Laser Cutter, 3D Printing
Programming: Python, Java, JavaScript, SwiftUI, Arduino, C++
Design: Figma, Photoshop, Adobe Illustrator
Languages: English, Spanish, Hebrew

ENGINEERING EXPERIENCE

Duke Motorsports Team, *Vehicle Dynamics* — August 2025-Present

- Collaborate with 20+ engineers to design, manufacture, and test a Formula SAE race car
- Led design of bearing carriers including CAD, load case analysis, and FEA, as well as expanding load cases analyzed by 100% and reducing weight by 15%
- Used Optimum Kinematics to evaluate suspension geometry and optimize camber, toe, and steering characteristics
- Designed and fabricated a 3D-printed steering sensor mount for integration with the steering rack
- Machine and manufacture parts using CNC mill, lathe, and 3D printers

Duke Co-Lab, *Makerspace Technician* — January 2026-Present

- Train students and researchers on equipment including 3D printers, laser cutters, waterjet, and ShopBot CNC systems
- Maintain 45+ 3D printers, perform troubleshooting, repairs, and preventative maintenance

Duke University Luck Biomechanics Lab, *Research Assistant* — January 2026-Present

- Conduct biomechanics experiments using a drop tower, load cell instrumentation, and anthropomorphic test devices
- Analyze and graph 600+ trials of gyroscopic acceleration data using Python

Duke University, *Software Engineering Intern* — May-July 2026

- Developed full-stack functionality for a research mobile application used by 100+ participants to collect wearable health data, building SwiftUI frontend interfaces and FastAPI backend services
- Built fitness dashboards, Duke Recreation event integration, social leaderboards, and in-app games to improve user engagement and support continuous health-data collection
- Presented project progress and final deliverables to researchers, stakeholders, and colleagues

Clemson University Department of Bioengineering, *Lab Aide Intern* — May-August 2024

- Created and passaged cell cultures used for testing
- Designed and developed an Arduino-controlled microscope stage using stepper motors and CAD-designed components

Army Educational Outreach Program (AEOP) Internship, *Lab Aide Intern* — June-August 2023

- Created cell cultures, ran experiments, and gathered data on cell signals with a flow cytometer
- Trained machine learning models on this data to recognize differences between cell signals
- Contributed to research published in *IEEE Sensors Journal* (DOI: 10.1109/JSEN.2025.3551914)

PROJECTS

Aquatic Knee Stabilization Device — August - December 2025

- Collaborated to create a knee brace that functions underwater to aid a spinal cord injury patient in physical therapy
- Used CAD and physical/mechanical/iterative design to engineer a working prototype

Microscope Stage Adjuster — May-August 2024

- Created a stage that can move precisely in 3 axes using Arduino and stepper motors

Personal Projects (see website)

- Steering Sensor Mount, CD Display Case, Happy Meal Toy, Sunflower Sheller, Toy RoboCop Helmet
- CAD projects demonstrating tolerance analysis, DFM principles, and iterative mechanical design

EXTRACURRICULAR EXPERIENCE

Duke STEM Connect — October 2025-Present

- Volunteer at a local Durham elementary school to design and teach fun and engaging STEM lessons
- Build relationships with 20+ students and encourage engagement with STEM subjects

INTERESTS

Basketball, cinema, Brazilian Jiu-Jitsu, music making, old school hip-hop